

Project Abstract: Student Bus Tracking system

Overview

A Student Bus Tracking system is a software and GPS-based tool used to track and manage the movement and location of buses used to transport students in real time. It improves communication, efficiency, and safety among parents, students, and schools. Students, parents, and school administrators can receive real-time updates on the whereabouts of school or college buses through this mobile application. It ensures students reach their destination safely and on time while keeping parents and administration informed.

User Roles

1.Admin

- Add/edit/delete buses, routes & stops .
- Register & manage users (drivers, students, parents) .
- Monitor bus locations in real-time.
- Generate reports (attendance, timings, delays).
- Handle complaints and feedback.

2.Student

- View bus location in real-time.
- Track estimated arrival time.
- Receive notifications (bus delay, cancellation).
- View route and stop info.
- Give feedback.

3.Driver

- Login and start assigned route.
- Update bus status (start, stop, delay, breakdown).
- View route, stops, and student list.
- Receive route changes or alerts.
- Mark student boarding/deboarding.

4.Parent

- Track child's bus in real-time.
 - Get alerts when the child boards or leaves the bus.
 - Receive emergency or delay notifications.
 - Access trip history.
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Core Functions

1.Real-time Bus Tracking

- Track the live location of buses using GPS.
- Display the location of the bus as a route map.
- Update location every few seconds.

2.Route management

- Create/edit/delete routes.
- Assign buses and drivers to specific routes.

3.Bus &Driver Management

- Add/edit/delete bus details.
- Maintain driver information.

4.Student enrollment &Stop mapping.

- Register students with assigned pickup/drop stops.
- Link students to specific bus routes.

5.ETA & Notification Alerts

- Show estimated arrival time to the student's stop.
- push notifications when Bus is near/Delays happen/Bus reaches stop.

Technology Stack

- Frontend:** HTML, CSS, JavaScript for user interface.
- Backend:** PHP for server side scripting.
- Database:** MySQL to store user information, bus details, routes, etc.
- Security:** Session handling and input validation.

Expected outcome

The aim of the Student Bus Tracking System is to give college or school buses a real-time tracking option. While drivers can update their location for parents or students can see the real-time location of their assigned bus on a root map. For every user the system offers secure login using session handling and input validation. Routes, buses, drivers, and student information can all be managed by admins. The expected outcome is a functional, user-friendly application with live tracking, route management, ETA display, and database-backed structure.